4. Consider the formula: $s = ut + \frac{1}{2}at^2$.
   (a) Which variable in this formula is the subject of the formula?
   (b) Find s given u = 5, t = 3, a = 6.
   (c) Find s given u = 10, t = 2, a = -3.

5. Consider the formula: $T = \sqrt{2p + 4b}$.
   (a) Find T given p = 1.5 and b = 5.5.
   (b) Find T given p = 4 and b = 14.

For the following, give answers correct to one decimal place where appropriate unless stated otherwise.

6. Consider the formula: $I = Prn$.
   (a) Find I given P = 1000, r = 0.05, n= 4.
   (b) Find I given P = 6000, r = 0.15, n = 5.

7. Consider the formula: $A = 2\pi rh$.
   (a) Find A given r = 9 and h = 6.
   (b) Find A given r = 5.6 and h = 15.

8. Consider the formula: $V = \frac{1}{3}\pi r^2 h$.
   (a) Find V given r = 4 and h = 12.
   (b) Find V given r = 10 and h = 15.

9. Consider the formula: $A = \pi(R^2 - r^2)$.
   (a) Find A given R = 12 and r = 7.
   (b) Find A given R = 20.5 and r = 12.5.

10. Consider the formula: $v = \sqrt{u^2 + 2as}$.
    (a) Find v given u = 4, a = 5, s = 3.
    (b) Find v given u = 14, a = -4, s = 12.5.

11. Consider the formula: $T = 2\pi\sqrt{\frac{L}{G}}$.
    (a) Find T given L = 50 and G = 32.
    (b) Find T given L = 30 and G = 28.

12. Consider the formula: $S = \frac{a(r^n - 1)}{r - 1}$.
    (a) Find S given a = 10, r = 0.9, n = 8.
    (b) Find S given a = 18, r = 0.75, n = 10.

13. Consider the formula: $A = P(1 + r)^t$.
    (a) Find A given P = 900, r = 0.15, t = 4.
    (b) Find A given P = 9900, r = 0.04, t = 6.

14. Consider the formula: $A = P(1+\frac{r}{n})^{nt}$.
 (a) Find A given P = 8008, r = 0.08, n = 4 and t = 5.
 (b) Find A given P = 20000, r = 0.06, n = 12 and t = 6.

15. The density of a body, D gm/cm^3 is given by the formula $D = \frac{M}{V}$ where M is the mass of the body in grams and V the volume in cm^3.
 (a) Find the density of gold if a ingot has a mass of 100 grams and a volume of 5.18 cm^3. Give your answer correct to 2 decimal places.

 (b) Find the density of iron if a mass of 1 kilograms has a volume of 128 cm^3.

 (c) Find the density of petrol if a 5 litre container of petrol is found to have a mass 3.5 kg.

 The density of pure water at 4°C is 1 gm/cm^3. Water at temperatures higher than 4°C has a density less than 1 gm/cm^3.
 (d) Find the density of water at 20°C gm/cm^3 if a mass of 1 kg has a volume of 1.002L. Give your answer correct to 3 decimal places.

16. "Young's Rule" is a way of calculating a child's medicine dose for children over the age of two. The formula for the rule is $C = \frac{nA}{n+12}$ where C is the child's dose, n is the child's age in years and A is the adult dose of the medicine in mL.
 If the adult dose of a particular medicine is 30mL find the dose for a child aged:
 (a) 8 years.
 (b) 30 months.

17. To calculate the total surface area of the human body (BSA) the Mostellar formula may be used.
 $BSA = \sqrt{\frac{W \times H}{3600}}\ m^2$ where W is the weight in kilograms and H the height in centimetres.
 (a) Find the BSA of a person of weight 72 kg and height 160 cm. Round answer to 2 decimal places.

 (b) Find the BSA of a person of weight 60 kg and height 1.5 m. Round answer to 2 decimal places.

18. The formula $C = \frac{pwt}{100\,000}$ gives the cost in dollars of running an electrical appliance, where p is the price of electricity in cents/kwH, w is the power that is needed to run the appliance in watts and t is the running time of the appliance in hours.
 Find the cost of running an electric motor rated at 1800 watts for 5 hours if the cost of electricity is 14.2 cents/kwH.

(c) 28.5 **4.** (a) s (b) 42 (c)) 14 **5.** (a) 5 (b) 8 **6.** (a) 200 (b) 4500 **7.** (a) 339.3 (b) 527.8 **8.** (a) 201.1 (b) 1570.8 **9.** (a) 298.5 (b) 829.4 **10.** (a) 6.8 (b) 9.8 **11.** (a) 7.9 (b) 6.5 **12.** (a) 57.0 (b) 67.9 **13.** (a) 1574.1 (b) 12526.7 **14.** (a) 11899.5 (b) 28640.9 **15.** (a) 19.31gm/cm^3 (b) 7.8125 gm/cm^3 (c) 0.7 gm/cm^3 (d) 0.998gm/cm^3 **16.** (a) 12 mL (b) 5.17 mL (2 d.p.) **17.** (a) 1.79 m^2 (b) 1.58 m^2 **18.** $1.28